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ADVANCED MICROECONOMICS

Consumer Theory I - Preferences and Utility

Winter Term 2025/2026

Consumer Theory

- Consumer theory model decisions of individual / households (HH).
- Producer theory models decisions of firms.
- Consumer theory **more important/central**.
 - In microeconomic theory, firms are ultimately owned by households.
 - Households provide the inputs (labor, capital, land)...
 - and they receive income in return (wages, interest, rent, profits).
 - Ultimate decision-making units in the economy are households, not firms.
- Decisions on household level **more complex**:
 - budget constraint(s) (time, income)
 - goals less straightforward than in case of firms (intertemporal, interpersonal)

Standard Approach - Rational Choice Analysis

- **Rational Choice:** HH have **rational preferences** which can be represented as **utility functions** and act according to them...
- HH take their decisions ...
 - ... so as to **maximize** their **utility** (captured by the utility function)
 - ... under a budget (and possibly other) constraint(s).
- **Conditions:** Consumers know which alternatives are available (complete information)
- Rational choice does not assume that people actually sit down and mathematically optimize their utility function, it just aims at describing the outcomes of human decisions **as if** they would mathematically optimize.

Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Comparative Statics (L4)
- 1.5) Duality, Slutsky Equation and Types of Goods (L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Safety Regulations and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

Overview

- 1 Introduction to Consumer Theory
- 2 Preferences
- 3 Utility functions and indifference curves
- 4 Marginal Rate of Substitution
- 5 Properties of Indifference Curves
- 6 Cardinal vs. Ordinal utility
- 7 Summary

Literature

- Varian (2022): Chapter 3+4 (*for refresher of basic concepts*)
- Autor (2016): Lecture Notes 3

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Preferences

- Preferences refer to consumers' ability to compare different bundles of goods.
- Mathematically this is expressed with **binary relations** - also called **preference relations**.
- Assume there are two bundles of goods (A, B) (where each bundle may contain multiple goods).
- Then there are different possible preference relations over these bundles:
 - $A \succsim B$ – consumer **prefers** A **weakly** over B (or “ A is at least as good as B ”)
 - $A \succ B$ – consumer **prefers** A **strictly** over B (or “ A is better than B ”, in LN $A^P B$)
 - $A \sim B$ – consumer is **indifferent** between A and B (or “ A is equal to B ”, in LN $A^I B$)

Axioms on preference relations

Overview

- **Axioms** are **basic assumptions** about preference relations
- Axioms provide the logical foundation on which we can build utility theory and demand analysis:
 - To allow for mathematical representation of utility functions
 - To portray rational (optimizing) behavior
 - To derive “well-behaved” demand curves
- **Everything** that follows (rational choice theory) rests entirely on these assumptions (That’s why its important to understand them!)

Axioms on Preference Relations

Core axioms (ensure utility representation)

- 1 Complete
- 2 Transitive
- 3 Continuous

→ If a preference relation fulfills these three axioms, we can always find a utility function that represents the preference relation.

Additional assumptions (make analysis easier)

- 4 Monotone (implies non-satiation)
- 5 Convex (implies diminishing marginal rate of substitution)

→ Axioms not necessary for finding utility functions but often intuitive (capture fundamental features of observed behavior) and furthermore useful because they make the maths easier.

Axiom 1 - Completeness

Definition

The preference relation $\succsim$ is complete if $\forall A, B \in X$ (where X is the set of all bundles of goods), $A \succsim B$ or $B \succsim A$ or both.

- The consumer can establish a preference ordering for any two bundles of goods A and B .
- For any comparison of bundles, the consumer will choose one and only one of the three options:
 - 1 $A \succ B$
 - 2 $B \succ A$
 - 3 $B \sim A$

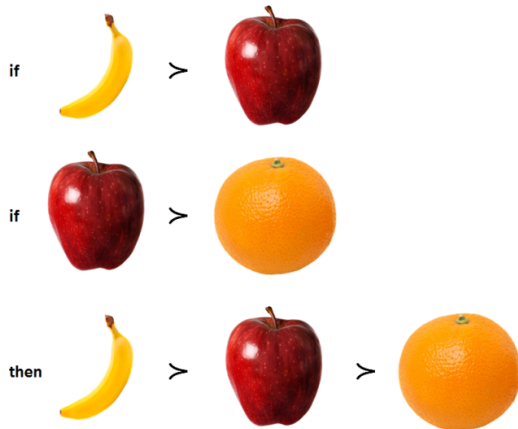
Axiom 2 - Transitivity

Definition

The preference relation $\succsim$ is **transitive**, if $A \succsim B$ and $B \succsim C$ implies $A \succsim C \forall A, B, C \in X$.

- Consumers are **consistent** in their preferences i.e., no circular preferences.
- **Example:**
 - Rolling a dice is transitive (6 beats 5, 5 beats 4, ... ,6 beats 1)
 - Rock–Paper–Scissors is not transitive (P beats R, R beats S but P does not S)
- Preferences that are both complete and transitive are called **rational** preferences.

Transitivity



Source: Paul Cohen - <https://paulcohen.com/rational-choice-axioms-visualized/>.

Axiom 3 - Continuity

Definition

The preference relation $\succsim$ is **continuous** if $A \succsim B$ and C lies in ϵ radius around B , then $A \succsim C \forall A, B, C \in X$.

- If you prefer bundle A to bundle B, then you won't suddenly flip your preference when we make small changes to B. $\rightarrow$ Preferences change “smoothly,” not in abrupt jumps.
- Not necessarily an intuitive property, but a property that is necessary to derive well-behaved demand curves.

Axiom 4 - Monotonicity (Non-satiation)

Definition

A and B each consist of two goods X and Y , where X_A is the quantity of X in bundle A etc.

If we assume that $Y_A = Y_B$ and $X_A > X_B$, a non-satiated preference relation implies that $A \succ B$ (strictly non-satiated) or $A \succeq B$ (weakly non-satiated).

- This axiom implies that **more is better** (strictly non-satiated) or **more cannot be worse** (weakly non-satiated).
- Consumer always places positive value on **more** consumption.

Axiom 5 - Convexity

Definition

The preference relation $\succsim$ is **convex** if for any two bundles of goods $A, B \in X$ where $B \succsim A$ and C a bundle on the line segment joining bundles A and B then $C \succsim A$ (weak convexity) or $C \succ A$ (strong convexity).

- In words: When B is at least as good as A , then every combination of A and B is at least as good as A (weak convexity) or better than A (strong convexity).
- Example: If I like to have 100 cups of coffee at least as much as I like 100 pieces of cake, I will like 50 cups of coffee with 50 pieces of cake more (or at least as much as) having only coffee or cake.

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Utility functions

- Preference relations are difficult to work with.
- If we are considering preferences that fulfill axioms 1-3 (completeness, transitivity and continuity), we can always find a **utility function** that **represents** the preference relation.
- Utility functions are mathematical functions which assign values (utility levels) to bundles of goods (e.g., $u(x, y) = x + y$, i.e., $u(A) = X_A + Y_A$).
- What does that mean?
 - A utility function $u()$ represents a preference relation $\succsim$ if $\forall A, B \in X, A \succsim B \Leftrightarrow u(A) \geq u(B)$.
 - Analogously also for other preference relations, i.e., $A \succ B \Leftrightarrow u(A) > u(B)$ and $A \sim B \Leftrightarrow u(A) = u(B)$.
 - Thus, the utility function preserves the **ranking of alternatives** of the preference relation.

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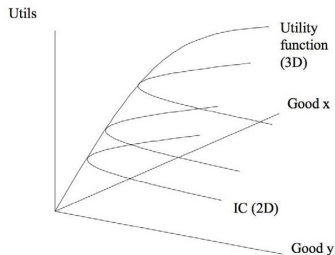
Indifference Curves

- Utility functions and preferences can be visualized using **indifference curves** (IDC).
- The map of all possible indifference curves represents a consumer's preferences.

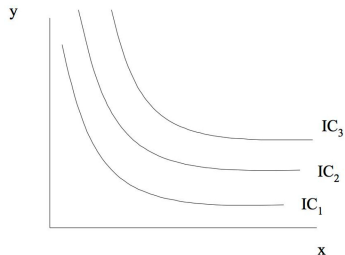
Indifference Curve

An indifference curve consists of all bundles of goods that give the consumer the **same utility level** $u(x) = \bar{u}$.

Indifference curves



Source: Autor (2016), Lecture Notes 3, p. 5



Source: Autor (2016), Lecture Notes 3, p. 5

Indifference curves and preferences

- With axioms 1–3, we can always find utility functions and indifference curves that represent the preferences.
 - With **Axiom 1 (completeness)**, every pair of consumption bundles can be compared, so every bundle lies on some indifference curve.
 - With **Axiom 2 (transitivity)**, indifference curves do not cross (preferences are consistent).
 - With **Axiom 3 (continuity)**, indifference curves are unbroken and connected (no jumps or gaps).
- With **Axiom 4 (non-satiation / monotonicity)**, there are infinitely many indifference curves and higher curves correspond to strictly preferred bundles.
- With **Axiom 5 (convexity)**, indifference curves are convex to the origin, which means a preference for variety (and implies a diminishing marginal rate of substitution).

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Marginal rate of substitution

Definition

The **marginal rate of substitution (MRS)** tells us how many units of good y the consumer is willing to give up to gain one additional unit of x .

- Mathematically this is the **slope of the indifference curve**.
- This slope may change along the indifference curve.

Marginal rate of substitution - Example

- **Assumptions**
 - 2 goods: hours of sleep (X) and points on the exam (Y)
 - 2 bundles: $A=(X = 5, Y = 60)$ and $B=(X = 6, Y = 50)$
- A and B are on one indifference curve (for student S) $\rightarrow$ S is indifference between A and B
- S is willing to give up 10 points in situation A to gain one additional hour of sleep.
- $\rightarrow$ $MRS(\text{sleep for points}) = \left| \frac{-10}{1} \right| = 10$
- S may be willing to give up only 5 points instead of 10 for one additional hour of sleep if the initial bundle had 7 instead of 5 hours of sleep $\rightarrow$ MRS depends on the starting point

Marginal rate of substitution and marginal utility

- Every point on the indifference curve has the same utility level,
 $u(x, y) = \bar{u}$
- When moving along the indifference curve, we know that $du = 0$
(no change in utility)
- We can take the **total derivative** of $u(x, y) = \bar{u}$:

$$0 = du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

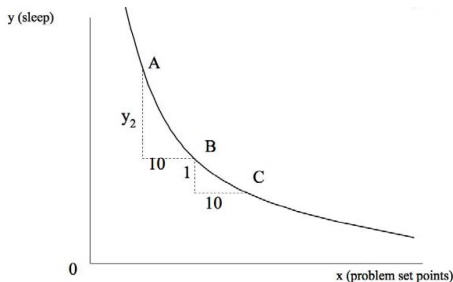
and rearrange

$$-\frac{dy}{dx} = \frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = \frac{MU_x}{MU_y} = MRS$$

- The MRS (of x for y) is thus equal to the ratio of the marginal utilities

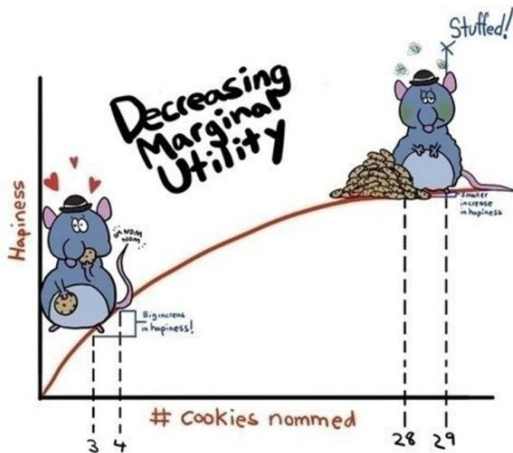
Diminishing marginal rate of substitution

- If MRS decreases when moving along the indifference curve, we call this a **diminishing marginal rate of substitution**
- I.e. if x is higher (and y lower, i.e., point B below instead of A) the consumer is willing to give up less y per additional unit of x :



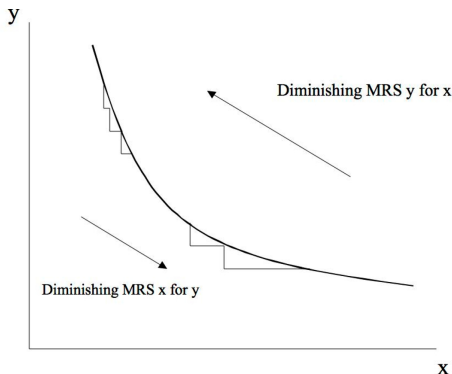
Source: Autor (2016), Lecture Notes 3, p. 8

Diminishing marginal utility



Source: Joy Buchanan, Economist writing Every Day (2023)

Diminishing marginal rate of substitution



Source: Autor (2016), Lecture Notes 3, p. 9

- MRS defined of x for y or the other way around
- Diminishing MRS implies that consumers prefer combinations of goods over extremes (only x or only y)

Overview

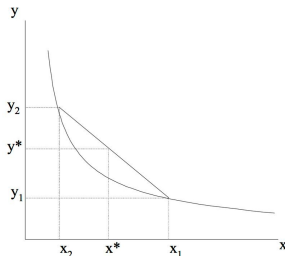
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Remember: Indifference curves and preferences

- With **Axiom 1 (completeness)**, every pair of consumption bundles can be compared, so every bundle lies on some indifference curve.
- With **Axiom 2 (transitivity)**, indifference curves do not cross (preferences are consistent).
- With **Axiom 3 (continuity)**, indifference curves are unbroken and connected (no jumps or gaps).
- With **Axiom 4 (non-satiation / monotonicity)**, there are infinitely many indifference curves and higher curves correspond to strictly preferred bundles.
- With **Axiom 5 (convexity)**, indifference curves are convex to the origin, which means a preference for variety (and implies a diminishing marginal rate of substitution).

Convexity

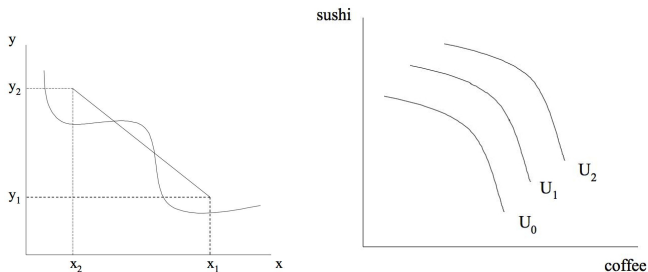
- A utility function has a diminishing marginal rate of substitution iff (if and only if) the IDC is convex.
- What is a convex function?
 - A function is convex if the line segment of any two points of the graph of the function lies above the graph of the function.



Source: Autor (2016), Lecture Notes 3, p. 9

Convexity

- Examples of non-convex curves - MRS is not (always) diminishing

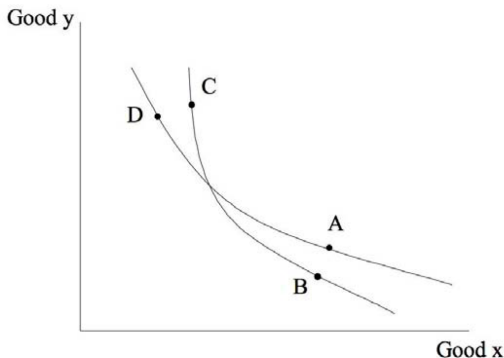


Source: Autor (2016), Lecture Notes 3, p. 10

- Remember: Convexity not a necessary condition for well-behaved demand curves $\rightarrow$ utility representation of these preferences is possible

Non-crossing indifference curves

- Transitivity (+ monotonicity) $\rightarrow$ non-crossing indifference curves



Source: Autor (2016), Lecture Notes 3, p. 11

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Cardinal vs. Ordinal utility

If a utility function $u()$ is to be interpreted **cardinally**, $u(A)$ has to represent a cardinal number (e.g. 'utils'), which implies that distances between different values of the function are meaningful.

- Example: If $u(A) = 12$ and $u(B) = 6$ an cardinal utility function, then A gives twice as much utility as B .
- Interpersonal comparisons possible

If a utility function is to be interpreted **ordinally**, it just keeps the ranking of the underlying preferences (the preference ordering) but the values are not themselves meaningful.

- Example: If $u(A) = 12$ and $u(B) = 6$, with an ordinal utility function A is preferred to B , but we cannot infer by how much A is better than B .
- No interpersonal comparisons possible

Ordinal utility in consumer theory

- For a **cardinal interpretation**, we'd have to **assume** that there are **absolute levels of utility**.
- Strong assumption **not needed** for most of consumer theory.
- To model decisions over bundles of goods an ordinal utility function (focus on the ranking/ordering of values) is sufficient.
- Important is only that the utility function keeps the ordering of alternatives and that the rate of substitution (MRS) is well-defined.

Monotonic Transformation

- **Problem:** How do we preserve existence of the MRS properties without imposing cardinality?
 - Even if different utility functions represent the same preferences, the MRS has to be the same for these different utility functions
- **Solution:** Weakened definition of utility functions → Utility function is only defined up to a **monotonic transformation**

Monotonic Transformation

A function $g(x)$ is a monotone transformation if it is a **strictly monotone increasing** function, i.e. it is a rank-preserving transformation.

Formally: If $g(x)$ is differentiable then $g'(x) > 0 \forall x$.

- Examples: $g(x) = x + 1$; $g(x) = 2x$; $g(x) = \ln(x)$ for $x > 0$

Monotonic Transformation of Utility Functions

If $U^2(x, y)$ is a monotone transformation of $U(x, y)$, i.e. $U^2(x, y) = f(U(x, y))$ where $f()$ is monotone in $U(x, y)$, then:

- ❶ U^1 and U^2 exhibit identical preference rankings
 - ❷ MRS of $U^1(\bar{U})$ and $U^2(\bar{U})$ are the same
 - ❸ U^1 and U^2 are equivalent for consumer theory
- A Monotonic transformation of a utility function is a utility function that represents the same preferences as the original function

Monotonic Transformation of Utility Functions

- Example:

① Cobb-Douglas utility function $U(x, y) = x^\alpha y^\beta$

$$\begin{aligned} dU &= \alpha x^{\alpha-1} y^\beta dx + x^\alpha \beta y^{\beta-1} dy = 0 \\ -\frac{dy}{dx} &= \frac{\alpha x^{\alpha-1} y^\beta}{x^\alpha \beta y^{\beta-1}} = \frac{\alpha}{\beta} \frac{y}{x} = MRS \end{aligned}$$

② Monotone transformation of $U(x, y)$, e.g.

$$U^2(x, y) = \ln(U(x, y)) = \alpha \ln x + \beta \ln y$$

$$\begin{aligned} dU^2 &= \frac{\alpha}{x} dx + \frac{\beta}{y} dy = 0 \\ -\frac{dy}{dx} &= \frac{\frac{\alpha}{x}}{\frac{\beta}{y}} = \frac{\alpha}{\beta} \frac{y}{x} = MRS \end{aligned}$$

Monotonic Transformation of Utility Functions

- General demonstration
 - Assume $U^2(x, y)$ is a monotone transformation of $U^1(x, y) = f(x, y)$, i.e., $U^2(x, y) = g(f(x, y))$.
 - Then MRS of $U^2(x, y)$:

$$\begin{aligned}
 0 &= g'(f(x, y)) \frac{\partial f}{\partial x} dx + g'(f(x, y)) \frac{\partial f}{\partial y} dy \\
 -\frac{dy}{dx} &= \frac{g'(f(x, y)) \frac{\partial f}{\partial x}}{g'(f(x, y)) \frac{\partial f}{\partial y}} \\
 &= \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = \frac{MU_x^1}{MU_y^1}
 \end{aligned}$$

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Summary

- ➊ Preferences are the foundation of decisions taken by consumers.
- ➋ Preferences can be expressed in form of preference relations.
- ➌ If preferences are rational (i.e., complete and transitive) and continuous, one can always find a utility function that represents the preferences (i.e., keeps the ordering of alternatives).
- ➍ Utility functions can be represented as indifference curves.
- ➎ Additional properties of preferences (axioms 4 and 5) buy additional (nice) properties of the utility functions and indifference curves.
- ➏ For consumer theory usually an ordinal interpretation of utility functions suffices.

Main Keywords

- Rational Choice
- Preference Relations
- Rationality
- Completeness
- Transitivity
- Continuity
- Monotonicity (Non-satiation)
- Convexity (Preference for Variety)
- Utility Function
- Indifference Curves
- Marginal Rate of Substitution (MRS)
- Ordinal vs. Cardinal Utility
- Monotonic Transformation

Questions?

